MUHAMMAD AINUL YAQIN

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Profile

Assistant Professor in the Department of Information Communication at Yuan Ze University. Ph.D. in Design from National Taipei University of Technology (2026). Specializes in HCI, UX research and assessment, and AIGC applications (NLP, RAG, VLM) for design innovation and education. More than five years of English-taught teaching and thesis mentoring, with publications in IEEE Transactions on Education and Interactive Learning Environments. Experienced in turning research into working systems, including AI agents and immersive VR learning tools.

Education

National Taipei University of Technology (NTUT) **2021 – 2026**
Doctoral Program in Design, College of Design, Taipei, Taiwan
GPA: 4.0/4.0

National Taipei University of Technology (NTUT) **2019 – 2021**
Master of Design, International Program of Interaction Design & Innovation, Taipei, Taiwan
GPA: 4.0/4.0

Taiwan Shoufu University (TSU) **2013 – 2017**
Bachelor of Science, Computer Science & Multimedia Design, Tainan, Taiwan
GPA: 4.0/4.0

Work Experience

Dept. of Information and Communication, YZU *Taoyuan, Taiwan* **Sept. 2026 – present**
↪ **Assistant Professor**

- Guide International student annually on thesis and research ideation, methods, and evaluation.
- Teach in the course of "Wearable Device Technologies and Speculative Design."

Dept. of Information and Communication, YZU *Taoyuan, Taiwan* **Feb. 2026 – Jul. 2026**
↪ **Lecturer**

- Teach in the course of "Exhibition Display Technology."

Smart Interaction & Service Design Laboratory, NTUT *Taipei, Taiwan* **Sept. 2021 – Jul. 2026**
↪ **Student Research Assistant**

- Guide 2–3 students annually on thesis ideation, methods, and evaluation.
- Assist ∼15 students/semester in UX evaluation and measurement topics.
- Collaborate with 10–15 lab members on innovative UX research directions and assessment.

Taiwan Architecture & Building Center (TABC) *New Taipei City, Taiwan* **Jul. 2024 – Nov. 2024**
↪ **Research Intern**

- Led a cross-functional team of 5 to design and implement an NLP-RAG-based AI Agent aligned with project goals.
- Managed end-to-end lifecycle: ideation, planning, execution, and deployment.

NTUT Extension Education Center *Taipei, Taiwan* **Feb. 2022 – Sept. 2023**
↪ **STEAM Educator for K-12**

- Taught classes of 10–15 students in Arduino & Makey Makey programming & hardware integration.
- Guided interactive projects and core coding concepts.

Digital Governance Association *Taipei, Taiwan* **Aug. 2021 – Feb. 2023**
↪ **Multimedia Specialist**

- Produced multimedia content for learning materials and podcasts.

Taione News Media Project *Taipei, Taiwan* **Oct. 2019 – Jun. 2020**
↪ **Multimedia Specialist**

- Executed multimedia projects with a focus on video editing.

PT. Win Access Telecommunication (VIVO) *Jakarta, Indonesia* **May 2019 – Aug. 2019**
↪ **Aftersales Management Officer**

- Standardized QA and improved customer satisfaction across Indonesia-West regional service centers.
- Translated technical documentation (Chinese to Bahasa Indonesia).

JU-SOFT Co., Ltd. *Taipei, Taiwan* **Sept. 2017 – Jul. 2018**
↪ **Product Application Specialist**

- Supported multimedia for live streaming & marketing; conducted field testing & technical support.

JU-SOFT Co., Ltd. *Taipei, Taiwan* **Mar. 2017 – Jun. 2017**
↪ **Multimedia Intern**

- Assisted multimedia content production, software testing, and IPTV promotional design.

Garuda Formosa TV *Taipei, Taiwan* **Mar. 2016 – Feb. 2017**
↪ **Broadcast Technician & Creative**

- Produced culturally tailored promotional/informational content and edited IPTV materials.

Academic Experience

Lectures

Designing Wearable Technology **Course Lecturer, Fall 2026**
Dept. Information Communication, Yuan Ze University

Speculative Design **Course Lecturer, Fall 2026**
Dept. Information Communication, Yuan Ze University

Exhibition Display Technology **Course Lecturer, Spring 2026**
Dept. Information Communication, Yuan Ze University

When Data Analysis Meets Design Thinking **Seminar Guest Speaker, Fall 2025**
Dept. of Computer Science, National Chengchi University (NCCU)

Human-computer Interaction Design **Lecturer, Spring 2025**
Dept. of Interaction Design, NTUT

Future Proof Portfolios: Leveraging GenAI for Professional Impact **Seminar Guest Speaker, Spring 2025**
Dept. Information Communication, Yuan Ze University

Design Methods: Prototyping for Designers **Seminar Guest Speaker, Spring 2022**
Dept. of Industrial Design, NTUT

Teaching Assistant

Design Thinking with Data Science **Fall 2025**
Dept. of Interaction Design, NTUT

Introduction to HCI **Fall 2023**
Dept. of Interaction Design, NTUT

Smart City & Interactivity **Fall 2022**
Dept. of Interaction Design, NTUT

Design Thinking **Fall 2021, 2024**
Dept. of Interaction Design, NTUT

VR Game Design (K-12 Camp) **Fall 2022**
NTUT Extension Education Center

Fundamental Electrical & Programming (K-12 Camp) **Spring, Fall 2022**
NTUT Extension Education Center

Arduino (K-12 Camp) **Spring, Fall 2022**
NTUT Extension Education Center

Thesis Mentoring

Jastine Joy Dela Paz **Graduated in 2026**
Designing Trustworthy Accessibility using Closed-Loop Adaptive Systems to Balance

User Agency and Transparency in AI-Mediated Web Interactions
Dept. Interaction Design, NTUT

Phan Minh Nan **Graduated in 2024**
Applying Online Text-based AI-Mediated Communication Methods for Mentor Seeking
Dept. Interaction Design, NTUT

Dang-My Linh **Graduated in 2024**
Promoting Financial Literacy Among Young Adults: A Narrative-Driven Serious Game Approach
Dept. Interaction Design, NTUT

Thanathep Thaithae **Graduated in 2024**
Cafescape: Personalized Café Soundscape Platform for Enhancing WFH
Dept. Interaction Design, NTUT

Kunal Prasad **Graduated in 2023**
SonicSurveyor: Urban Noise Soundscape Mapping
Dept. Interaction Design, NTUT

Alex M. Chong **Graduated in 2022**
Self-Regulating Traction Towards Pursuits Through Awareness Prompting
Dept. Interaction Design, NTUT

Publications

[1] S.-M. Wang, M. A. Yaqin, and M. N. Phan, "Optimizing mentor-mentee matching: a framework for ai-mc as a ux approach in online learning," *Interactive Learning Environments*, pp. 1–20, 2025.

[2] S.-M. Wang, M. A. Yaqin, and V. H. Lan, "Enhancing spatial-reasoning perception using virtual reality immersive experience," *IEEE Transactions on Education*, vol. 67, no. 5, pp. 648–659, 2024.

[3] D. Kozachek, M. A. Yaqin, K. Prasad, and S.-M. Wang, "Evaluating the outcome of collaborative vr mind mapping sessions with sentiment analysis and emotional intelligence," in *Human-Computer Interaction* (M. Kurosu and A. Hashizume, eds.), (Cham), pp. 245–263, Springer Nature Switzerland, 2023.

[4] S.-M. Wang, M. A. Yaqin, and F.-H. Hsu, "Ahp-based assessment of developing online virtual reality services with progressive web apps," in *New Trends in Computer Technologies and Applications* (S.-Y. Hsieh, L.-J. Hung, R. Klasing, C.-W. Lee, and S.-L. Peng, eds.), (Singapore), pp. 553–566, Springer Nature Singapore, 2022.

[5] S.-M. Wang, M. A. Yaqin, and F.-H. Hsu, "Improving emotional intelligence in the new normal using metaverse applications for digital native," in *Human-Computer Interaction. Theoretical Approaches and Design Methods* (M. Kurosu, ed.), (Cham), pp. 527–541, Springer International Publishing, 2022.

Projects

Research Projects

Human-in-the-Loop Visual Language Models for Responsible Artificial Intelligence Supported Visual Design Reasoning: A Multimodal Mixed-Methods Study **Sept. 2026 – Jul. 2027**
PI: Assoc. Professor Sheng-Ming Wang - NTUT, Assist. Professor Chieh-Ju Huang - YZU, and Assist. Professor Muhammad A. Yaqin - YZU

The Development of Industry Knowledge Databases for AI Design Applications **Sept. 2026 – Nov. 2026**
PI: Assoc. Professor Sheng-Ming Wang - NTUT, Assist. Professor Chieh-Ju Huang - YZU, and Assist. Professor Muhammad A. Yaqin - YZU

Integrating Retrieval Augmented Generation and Scaffolding Theory to Enhance Computational Thinking and Technology Capability in Human-Computer Interaction Design Practice **Aug. 2025 – July 2026**
Advisor: Dr. Sheng-Ming Wang - Assoc. Professor, Dept. of Interaction Design, NTUT

Venture Your Green Investment Capital: AI REITs Platform **Jul. 2024 – Nov. 2024**
Advisor: Morris Lin - Manager, Dept. of Digital Twin for Sustainable City, TABC

Integrating Big and Thick Data to Fine-Tune Open-Source Large Language Models for Urban Disaster Governance and the Operation of Smart Spatial Information Systems **Feb. 2023**
Advisor: Dr. Sheng-Ming Wang - Assoc. Professor, Dept. of Interaction Design, NTUT

Enhancing interventions and risk prediction models for children with developmental delays through artificial intelligence to improve early diagnosis and early intervention rates

in rural and indigenous areas — AI-based automatic detection and assessment of children's gross and fine motor development. **Sept.– Feb. 2023, Oct.– Dec. 2025**
Advisor: Dr. Tun-Wen Pai - Professor, Dept. of Computer Science and Information Engineering, NTUT

A Multimodal Data-Driven Analysis Framework for Immersive Collaborative Learning in Virtual Reality **Nov. 2022**
Advisor: Dr. Sheng-Ming Wang - Assoc. Professor, Dept. of Interaction Design, NTUT

Evaluating the Outcome of Collaborative VR Mind Mapping Sessions with Sentiment Analysis and Emotional Intelligence **Jul. 2022 – Sep. 2022**
Advisor: Dr. Sheng-Ming Wang - Assoc. Professor, Dept. of Interaction Design, NTUT

Evaluating Collaborative Interaction and Performance in Virtual Reality Application with Multi-Datasets Correlative Analysis (*A collaboration project with FU Berlin)* **Jul. 2022 – Sep. 2022**
Advisor: Dr. Sheng-Ming Wang - Assoc. Professor, Dept. of Interaction Design, NTUT

The Integration of Kano Model and DEA for Evaluating the Immersive Experience in Web Extended Reality Innovative Service **Nov. 2020 – Jul. 2021**
Advisor: Dr. Sheng-Ming Wang - Assoc. Professor, Dept. of Interaction Design, NTUT

Speculative Design for Self-Health Monitoring Patch **Feb. 2020 – Jun. 2020**
Advisor: Dr. Sheng-Ming Wang - Assoc. Professor, Dept. of Interaction Design, NTUT

Skills & Interests

Technical	Python, C++, HTML, CSS, Scratch, SPSS, JASP, Adobe Family, Arduino, Makey-makey.
Design Research	Data Analytics, UX Research, Prototyping, UI Design, Design Thinking, Service Design, Speculative Design, Post-video Editing, TouchDesigner.
Languages	Indonesian (Native), English (Professional), Chinese (Limited).
Research Interests	UX Research & Assessment, HCI, Service Design, AIGC Application (NLP & VLM), GUI, STEAM, Digital Technology & Education.
Course Alignment	HCID, HCAI Applications, UX Research, Research Design and Data Acknowledgments, Wearable Design Technology, Speculative Design, AI Interactive Applications and related fields.

Awards & Scholarships

Bronze Award **2024**
Global Digital Talent Awards & Digital Equality Forum

Selected Awardee **2024**
2024 Taiwan DIGI+ & Talent Circulation Alliance Internship Program

Hult Prize Taiwan - on Campus Program Winner **2022**
Hult Prize Foundation in Associate with United Nations

International Graduate Student Scholarship Awardee (Ph.D) **2021 – 2025**
National Taipei University of Technology

Top 3 Best Graduate Student (Master) in College of Design **2021**
National Taipei University of Technology

Best Academic Paper Awards **2020**
International Conference on Medical Design, Taoyuan, Taiwan

International Graduate Student Scholarship Awardee (Master) **2019 – 2021**
National Taipei University of Technology

Taipei City ICT Southbound Diamond Talent Convergence Plan Awardee **2019 – 2021**
Taipei City Government, Taiwan

Top 10 Best Graduate Student (Bachelor) **2017**
Taiwan Shoufu University

2nd Winner Micro-Video School Competition **2016**
Taiwan Shoufu University

Best Performance of 2015 Asian Golden Awards of "Film-Micro Contest" **2015**
Asia Golden Dragon Award, Taiwan

International Undergraduate Student Scholarship Awardee **2013 – 2017**
Taiwan Shoufu University